Ofqual QAN	**600/3055/0**
Quartz ID	**1158 / 1159 / 1161**
Qualification	**Level 5 Diploma in Computing / Level 5 Diploma in Computing with Business Management / Level 5 Diploma in Computing with Data Science**

Assessment	**Introduction to Data Science and Big Data**
Assessment Type	**Examination**
Date	**Wednesday 28th May 2025**
Time	**9.00am – 12.00pm**

Examination Paper

Answer ALL questions.

Clearly cross out surplus answers.

Time: 3 Hours

The maximum mark for this paper is 100.

Any reference material brought into the examination room must be handed to the invigilator before the start of the examination.

Answer ALL questions

Marks

Question 1

a) Discuss the differences between a Data Scientist, a Data Analyst and a Big Data Expert, focusing on their roles **and** responsibilities. **6**

b) Select **TWO (2)** of the roles in part (a). For each role highlight unique key skills and explain how these skills contribute to their work. **4**

Total 10 Marks

Question 2

a) Identify **and** describe **THREE (3)** main sources of data in the modern digital world **and** provide **ONE (1)** example for each. **6**

b) Discuss **TWO (2)** challenges associated with data quality **and** their potential solutions. **4**

Total 10 Marks

Question 3

a) List **and** define the **SIX (6) broad categories** of data types used in data science. **6**

b) Explain how understanding data types is essential in data preprocessing and analysis. **4**

Total 10 Marks

Question 4

a) Define Central Tendency **and** explain its role in Exploratory Data Analysis (EDA). **4**

b) Provide examples of **THREE (3) measures** of central tendency, explain their definitions **and** describe their applications. **6**

Total 10 Marks

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Marks

Question 5

a) What is binning in the context of data preprocessing? **7**

Explain the difference between **THREE (3)** different binning techniques with an example of each.

b) Define class imbalance in datasets. **3**

Explain why class imbalance is a challenge for machine learning models?

Total 10 Marks

Question 6

a) Explain the concept of Reinforcement Learning (RL) **and** discuss how it differs from Supervised Learning. **5**

b) Provide **ONE (1)** real-world example where Reinforcement Learning is applied, explaining how it improves decision-making. **5**

Total 10 Marks

Question 7

a) Describe the **THREE (3)** phases of understanding in data visualisation: perceiving, interpreting and comprehending. **6**

b) Explain how the phases contribute to deriving meaningful insights from data. **4**

Total 10 Marks

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Marks

Question 8

a) Define Mean Absolute Error (MAE) **and** explain its importance in regression analysis. **4**

b) Define AUC (Area Under the ROC Curve) **and** explain its importance in classification problem evaluation. **6**

Total 10 Marks

Question 9

a) Discuss **TWO (2)** components of Business Intelligence architecture: Data Warehousing and Business Analytics. Use examples to illustrate your answer. **5**

b) Explain how Data Warehousing **and** Business Analytics contribute to informed decision-making, giving examples. **5**

Total 10 Marks

Question 10

a) Explain the concepts of Privacy, Fairness and Transparency in the context of Data Ethics. Use examples to illustrate your answer. **5**

b) Explain why maintaining fairness and transparency is critical in data-driven decision-making. **5**

Provide an example where a lack of these aspects leads to ethical concerns.

Total 10 Marks

End of paper

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